

Please complete all of the following problems.

Show all your work.

6.1 Why do heat (q) and work (w) have positive values when entering a system and negative values when leaving?

6.6 In winter, an electric heater uses a certain amount of electrical energy to heat a room to 20°C . In summer, an air conditioner uses the same amount of electrical energy to cool the room to 20°C . Is the change in internal energy of the heater larger, smaller, or the same as that of the air conditioner? Explain.

6.8 A system receives 425 J of heat and delivers 425 J of work to its surroundings. What is the change in internal energy of the system (in J)?

6.10 What is the change in internal energy (in J) of a system that releases 675 J of thermal energy to its surroundings and has 525 cal of work done on it?

6.12 Complete combustion of 1.0 metric ton of coal (assuming pure carbon) to gaseous carbon dioxide releases 3.3×10^{10} J of heat. Convert this energy to (a) kilojoules; (b) kilocalories; (c) British thermal units.

(1 BTU is approximately 1055 J)

6.14 The nutritional calorie (Calorie) is equivalent to 1 kcal. One pound of body fat is equivalent to about 4.1×10^3 Calories. Express this quantity of energy in joules and kilojoules.

6.15 If an athlete expends 1850 kJ/h, how long does she have to play to work off 1.0 lb of body fat? (See Problem 6.14.)

6.17 Why is it usually more convenient to measure ΔH than ΔE ?

6.19 Classify the following processes as exothermic or endothermic: (a) freezing of water; (b) boiling of water; (c) digestion of food; (d) a person running; (e) a person growing; (f) wood being chopped; (g) heating with a furnace.

6.22 Draw an enthalpy diagram for a general exothermic reaction; label axis, reactants, products, and ΔH with its sign.

6.24 Write a balanced equation and draw an approximate enthalpy diagram for each of the following: (a) the combustion of 1 mol of methane in oxygen; (b) the freezing of liquid water.

6.33 Calculate q when 12.0 g of water is heated from $20.^{\circ}\text{C}$ to $100.^{\circ}\text{C}$.

6.35 A 295-g aluminum engine part at an initial temperature of 3.00°C absorbs 85.0 kJ of heat. What is the final temperature of the part (c of Al = $0.900 \text{ J/g}\cdot\text{K}$)?

6.37 Two iron bolts of equal mass—one at $100.^{\circ}\text{C}$, the other at 55°C —are placed in an insulated container. Assuming the heat capacity of the container is negligible, what is the final temperature inside the container (c of iron = $0.450 \text{ J/g}\cdot\text{K}$)?

6.39 When 165 mL of water at 22°C is mixed with 85 mL of water at 82°C , what is the final temperature? (Assume that no heat is lost to the surroundings; d of water is 1.00 g/mL .)

6.41 A 505-g piece of copper tubing is heated to 99.9°C and placed in an insulated vessel containing 59.8 g of water at 24.8°C . Assuming no loss of water and a heat capacity for the vessel of 10.0 J/K , what is the final temperature of the system (c of copper = $0.387 \text{ J/g}\cdot\text{K}$)?

6.48 Would you expect $\text{O}_2(\text{g}) \longrightarrow 2\text{O}(\text{g})$ to have a positive or a negative ΔH_{rxn} ? Explain.

6.49 Is ΔH positive or negative when 1 mol of water vapor condenses to liquid water? Why? How does this value compare with that for the vaporization of 2 mol of liquid water to water vapor?

6.50 Consider the following balanced thermochemical equation for a reaction sometimes used for H_2S production:



(a) Is this an exothermic or endothermic reaction?

(b) What is ΔH_{rxn} for the reverse reaction?

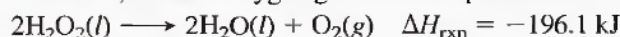
(c) What is ΔH when 3.2 mol of S_8 reacts?

(d) What is ΔH when 20.0 g of S_8 reacts?

(remember that a balanced thermochemical equation can have fractional coefficients, as in this example, we are concerned with change in enthalpy for each mole of *hydrogen sulfide* produced.)

6.52 When 1 mol of $\text{NO}(\text{g})$ forms from its elements, 90.29 kJ of heat is absorbed. (a) Write a balanced thermochemical equation for this reaction. (b) How much heat is involved when 1.50 g of NO decomposes to its elements?

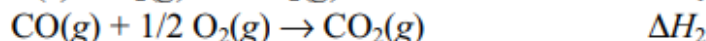
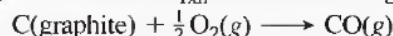
6.54 Liquid hydrogen peroxide, an oxidizing agent in many rocket fuel mixtures, releases oxygen gas on decomposition:



6.58 Ethylene (C_2H_4) is the starting material for the preparation of polyethylene. Although typically made during the processing of petroleum, ethylene occurs naturally as a fruit-ripening hormone and as a component of natural gas.

(a) If the heat of combustion of C_2H_4 is -1411 kJ/mol , write a balanced thermochemical equation for the combustion of C_2H_4 .
(b) How many grams of C_2H_4 must burn to give 70.0 kJ of heat?

6.62 It is very difficult to burn carbon in a deficiency of O_2 and produce only CO; some CO_2 forms as well. However, carbon burns in excess O_2 to form only CO_2 , and CO burns in excess O_2 to form only CO_2 . Use the heats of the latter two reactions (from Appendix B) to calculate ΔH_{rxn} for the following reaction:



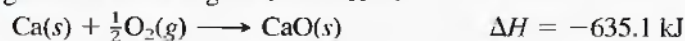
Where $\Delta H_1 = -393.5 \text{ kJ/mol}$ carbon dioxide

and $\Delta H_2 = -283.0 \text{ kJ/mol}$ carbon dioxide (from Appendix B)

6.63 Calculate ΔH_{rxn} for

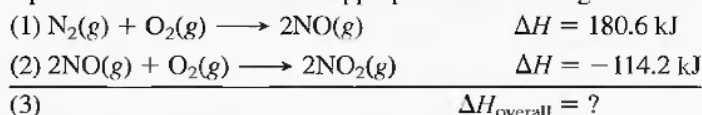


given the following set of reactions:



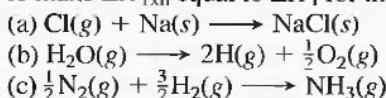
6.65 At a specific set of conditions, 241.8 kJ is given off when 1 mol of $\text{H}_2\text{O}(\text{g})$ forms from its elements. Under the same conditions, 285.8 kJ is given off when 1 mol of $\text{H}_2\text{O}(\text{l})$ forms from its elements. Calculate the heat of vaporization of water at these conditions.

6.67 Write the balanced overall equation for the following process (equation 3), calculate $\Delta H_{\text{overall}}$, and match the number of each equation with the letter of the appropriate arrow in Figure P6.67:



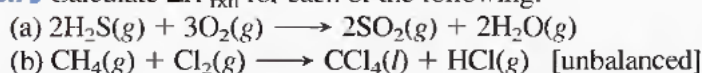
6.70 What is the difference between the standard heat of formation and the standard heat of reaction?

6.72 Make any changes needed in each of the following equations to make ΔH_{rxn}^0 equal to ΔH_f^0 for the compound present:

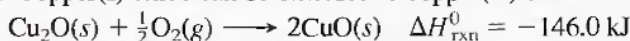


6.73 Use Table 6.3 or Appendix B to write balanced formation equations at standard conditions for each of the following compounds: (a) CaCl_2 ; (b) NaHCO_3 ; (c) CCl_4 ; (d) HNO_3 .

6.75 Calculate ΔH_{rxn}^0 for each of the following:



6.77 Copper(I) oxide can be oxidized to copper(II) oxide:



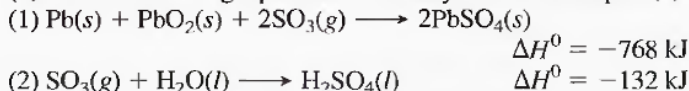
Given that ΔH_f^0 of $\text{Cu}_2\text{O}(\text{s}) = -168.6 \text{ kJ/mol}$, what is ΔH_f^0 of $\text{CuO}(\text{s})$?

6.80 The common lead-acid car battery produces a large burst of current, even at low temperatures, and is rechargeable. The reaction that occurs while recharging a “dead” battery is



(a) Use ΔH_f^0 values from Appendix B to calculate ΔH_{rxn}^0 .

(b) Use the following equations to check your answer in part (a):



6.81 Stearic acid ($\text{C}_{18}\text{H}_{36}\text{O}_2$) is a typical fatty acid, a molecule with a long hydrocarbon chain and an organic acid group (COOH) at the end. It is used to make cosmetics, ointments, soaps, and candles and is found in animal tissue as part of many saturated fats. In fact, when you eat meat, chances are that you are ingesting some fats that contain stearic acid.

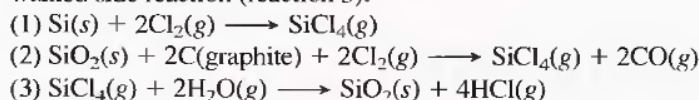
(a) Write a balanced equation for the complete combustion of stearic acid to gaseous products.

(b) Calculate its heat of combustion ($\Delta H_f^0 = -948 \text{ kJ/mol}$).

(c) Calculate the heat (q) in kJ and kcal when 1.00 g of stearic acid is burned completely.

(d) The nutritional information for a candy bar states that one serving contains 11.0 g of fat and 100. Cal from fat (1 Cal = 1 kcal). Is this information consistent with your answer for part (c)?

6.105 Silicon tetrachloride is produced annually on the multikiloton scale for making transistor-grade silicon. It can be made directly from the elements (reaction 1) or, more cheaply, by heating sand and graphite with chlorine gas (reaction 2). If water is present in reaction 2, some tetrachloride may be lost in an unwanted side reaction (reaction 3):



$$\Delta H_{\text{rxn}}^0 = -139.5 \text{ kJ}$$

(a) Use reaction 3 to calculate the heats of reaction of reactions 1 and 2. (b) What is the heat of reaction for the new reaction that is the sum of reactions 2 and 3?

Table 6.3 Selected Standard Heats of Formation at 25°C (298 K)

Formula	ΔH_f^0 (kJ/mol)
Calcium	
Ca(s)	0
CaO(s)	-635.1
CaCO ₃ (s)	-1206.9
Carbon	
C(graphite)	0
C(diamond)	1.9
CO(g)	-110.5
CO ₂ (g)	-393.5
CH ₄ (g)	-74.9
CH ₃ OH(l)	-238.6
HCN(g)	135
CS ₂ (l)	87.9
Chlorine	
Cl(g)	121.0
Cl ₂ (g)	0
HCl(g)	-92.3
Hydrogen	
H(g)	218.0
H ₂ (g)	0
Nitrogen	
N ₂ (g)	0
NH ₃ (g)	-45.9
NO(g)	90.3
Oxygen	
O ₂ (g)	0
O ₃ (g)	143
H ₂ O(g)	-241.8
H ₂ O(l)	-285.8
Silver	
Ag(s)	0
AgCl(s)	-127.0
Sodium	
Na(s)	0
Na(g)	107.8
NaCl(s)	-411.1
Sulfur	
S ₈ (rhombic)	0
S ₈ (monoclinic)	0.3
SO ₂ (g)	-296.8
SO ₃ (g)	-396.0